Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2024-P2-Model) | Date: 26-10-2025

Time: 02.00Pm to 05.00Pm | GTA-04 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**C**	2	**B**	3	**D**	4	**C**	5	**ABCD**	6	**AC**
7	**ABC**	8	**16**	9	**5**	10	**88**	11	**95**	12	**60**
13	**1**	14	**48**	15	**31**	16	**0.83**	17	**0**		

PHYSICS

18	**C**	19	**C**	20	**D**	21	**D**	22	**AC**	23	**ABCD**
24	**AC**	25	**32**	26	**6**	27	**4**	28	**8**	29	**4**
30	**2**	31	**666.65-666.70**	32	**3.10**	33	**1.5**	34	**0.30**		

CHEMISTRY

35	**C**	36	**B**	37	**D**	38	**B**	39	**ABD**	40	**ABC**
41	**ABCD**	42	**2**	43	**3**	44	**4**	45	**4**	46	**3**
47	**2**	48	**1.2**	49	**0.6**	50	**0.50**	51	**6.8**		

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. $\sum_{r=0}^{22}(-1)^r \frac{^{20}C_r}{^{r+2}C_2} = \sum_{r=0}^{20}(-1)^r \frac{20!}{r!(20-r)!} \times \frac{2!r!}{(r+2)!}$

$= \frac{2}{21\times 22}\sum_{r=0}^{20}(-1)^r\, {}^{22}C_{r+2}$

$= \frac{2}{21\times 22}\left[{}^{22}C_2 - {}^{22}C_3 + \ldots {}^{22}C_{22} + {}^{22}C_0 - {}^{22}C_1 - \left({}^{22}C_0 - {}^{22}C_1\right)\right]$

$= \frac{2}{21\times 22}\left[(1-1)^{22} - (1-22)\right] = \frac{1}{11}$

2. Co-ordinate of A_n is $\left(n^2 - n, \frac{\sqrt{3}}{2}(2n-1)\right)$

Eliminating n, locus equation is $y^2 = 3\left(x + \frac{1}{4}\right)$

3. $f(x) = \left(\frac{3x}{\sqrt{x^2+1}} + \frac{\sqrt{x^2+1}}{x}\right) + \left(\frac{x^2+1}{\sqrt{x^4+2}} + \frac{\sqrt{x^4+2}}{x^2+1}\right)$

Minimum value is attained when, $\frac{3x}{\sqrt{x^2+1}} + \frac{\sqrt{x^2+1}}{x}$ *i.e* $x = \frac{1}{\sqrt{2}}$.

4. $\prod_{k=0}^{97}\left(2 + 3\alpha_k^2\right) = \prod_{k=0}^{97}\left(\sqrt{3}\alpha_k + i\sqrt{2}\right)\left(\sqrt{3}\alpha_k - i\sqrt{2}\right)$

$= 3^{98}\prod_{k=0}^{97}\left(\frac{-i\sqrt{2}}{\sqrt{3}} - \alpha_k\right)\prod_{k=0}^{97}\left(\frac{i\sqrt{2}}{\sqrt{3}} - \alpha_k\right) = 3^{98}\left(\left(\frac{-i\sqrt{2}}{\sqrt{3}}\right)^{98} - 1\right)\left(\left(\frac{i\sqrt{2}}{\sqrt{3}}\right)^{98} - 1\right) = \left(2^{49} + 3^{49}\right)^2$

5. $pA^2 = p^2A + A^3$

$\frac{p^2A = p^3I + pA^2}{A^3 = p^3I = 0}, A^3 = -p^3I$

$|A|^3 = \left(-p^3\right)^3, |A| = -p^3$

6. Let, $x = p + q$

$\Rightarrow y = p + r$ *then*, $\frac{1}{p+q} + \frac{1}{p+r} = \frac{1}{p}$

$\Rightarrow p^2 = qr$

q	r
1	p^2
p	p
p^2	1

7. The given hyperbola is a rectangular hyperbola so ΔCEF is right angle and eccentricity of the hyperbola is $\sqrt{2}$

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8. $\lim\limits_{a\to\infty}\dfrac{1}{a}\int\limits_0^{\infty}\dfrac{x^2+ax+1}{1+x^4}\tan^{-1}\left(\dfrac{1}{x}\right)dx$

$=\lim\limits_{a\to\infty}\dfrac{1}{a}\left(\int\limits_0^{\infty}\dfrac{x^2+1}{1+x^4}\tan^{-1}\left(\dfrac{1}{x}\right)dx+a\int\limits_0^{\infty}\dfrac{x}{1+x^4}\tan^{-1}\left(\dfrac{1}{x}\right)dx\right)$

$I_1=\int\limits_0^{\infty}\dfrac{x^2+1}{1+x^4}\tan^{-1}\left(\dfrac{1}{x}\right)dx \quad I_2=\int\limits_0^{\infty}\dfrac{x}{1+x^4}\tan^{-1}\left(\dfrac{1}{x}\right)dx$

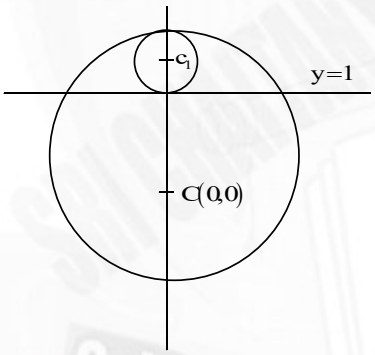
$x=\dfrac{1}{t},\ dx=\dfrac{-1}{t^2}dt \quad 2I_2=\int\limits_0^{\infty}\dfrac{t}{t^4+I}\left(\tan^{-1}\left(\dfrac{1}{t}\right)+\tan^{-1}(t)\right)dt$

$I_2=\dfrac{\pi}{4}\left[\int\limits_0^{\infty}\dfrac{\frac{d\lambda}{2}}{\lambda^2+1}\right]=\dfrac{\pi}{8}\tan^{-1}(\lambda)\Big|_0^{\infty}=\dfrac{\pi^2}{16}$

$I_1=\dfrac{\pi}{4}\int\limits_0^{\infty}\dfrac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt=\dfrac{\pi}{4}\int\limits_{-\infty}^{\infty}\dfrac{du}{u^2+2}\left(u=t-\dfrac{1}{t}\right)=\dfrac{\pi^2}{4\sqrt{2}}$

$\lim\limits_{a\to0}\dfrac{1}{a}\left[\dfrac{\pi}{4\sqrt{2}}+a\left(\dfrac{\pi^2}{16}\right)\right]=\dfrac{\pi^2}{16}.$

9.

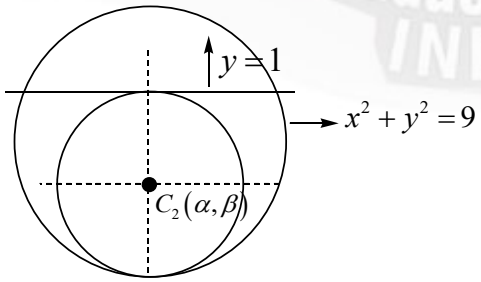


$CC_1=3-R$

$\alpha^2+\beta^2=(3-R)^2,(\beta-1)=R\Rightarrow\beta=1+R$

$\alpha^2=(3-R)^2-(1+R)^2,=(9+R^2-6R)-(R^2+1+2R)$

$\alpha^2=8-8R\Rightarrow R=1$ then centre lies on y-axis



$CC_1=3-R_2$

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$1-\beta=R\Rightarrow\beta=1-R$

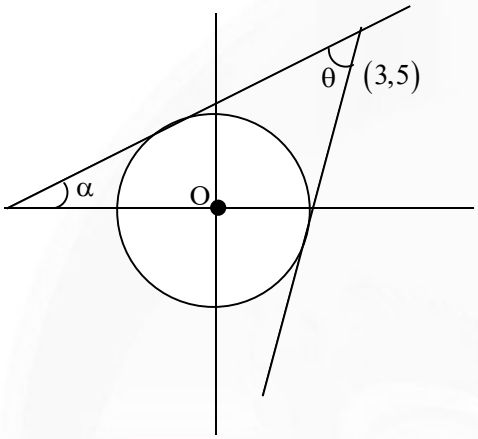
$\alpha^2+\beta^2=(3-R)^2$

$\Rightarrow\alpha^2=(3-R)^2-(1-R)^2=8-4R$

$\Rightarrow R=2$ then centre lies on y-axis

10. Focus of parabola is $(3,5)$ and focal chord is

$mx-y+5-3m=0$



$\therefore\left|\frac{5-3m}{\sqrt{1+m^2}}\right|=3\Rightarrow m=\frac{8}{15}\ or\ \infty\ \therefore\tan\alpha=\frac{8}{15}\Rightarrow\theta=\frac{\pi}{2}-\alpha=\cot^{-1}\left(\frac{8}{15}\right)=\tan^{-1}\left(\frac{15}{8}\right)$

Maximum number of functions formed when A contains 4 elements and B contains 3 elements.

11. $f(z)=z^2-19z$ $\quad\frac{ff(z)-f(z)}{z-f(z)}$ is imaginary

$\frac{(z^2-19z)^2-19(z^2-19z)-(z^2-19z)}{z-(z^2-19z)}$

$\frac{(z^2-19z)^2(z^2-19z-20)}{(20z-z^2)}z^2-20z+z-20$

$-(z-19)(z+1)$ is imaginary $\quad-(z^2-18z-19)$

$-\left((a+11i)^2-18(z+11i)-19\right)$

$-(a^2-121+22ai-18a-198i-19)$

$-(a^2)-18z+140+i(22a-198))$ is imaginary

$a^2-18z-140=0\quad a=\frac{18\pm\sqrt{324}+4\times140}{2}\quad=9\pm\sqrt{221}$

$|AB|=0\ as\ AB=\begin{bmatrix}x+1 & x+2 & 0\\ 3x+3 & x+5 & 0\\ 6x+2 & x+4 & 0\end{bmatrix}\begin{bmatrix}x+1 & x+2 & 5x+7\\ 3x+3 & x+5 & 8x\\ 6x+2 & x+4 & 0\end{bmatrix}$

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12. $.\ {}^{6}C_{5}.5!\left(44+{}^{5}C_{1}\left(9+44\right)\right)=\left(309\right)6!$

13. $A=\sum_{r=1}^{3}\log\left|\tan\left(60^{0}-\alpha_{r}\right)\right|$

$B=\sum_{r=1}^{3}\log\left|\tan\left(60^{0}+\alpha_{r}\right)\right|$

$A+B=\log_{(2+\sqrt{3})}\left|\tan\left(60-\alpha_{1}\right)\tan\left(60+\alpha_{1}\right)\right|$

$+\log_{(2+\sqrt{3})}\left|\tan\left(60-\alpha_{2}\right)\tan\left(60+\alpha_{2}\right)\right|$

$+\log_{(2+\sqrt{3})}\left|\tan\left(60-\alpha_{3}\right)\tan\left(60+\alpha_{3}\right)\right|$

$A+B=\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(3\alpha_{1}\right)}{\tan\left(\alpha_{1}\right)}\right|+\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(3\alpha_{2}\right)}{\tan\left(\alpha_{2}\right)}\right|+\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(3\alpha_{3}\right)}{\tan\left(\alpha_{3}\right)}\right|$

$A+B=\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(3\alpha_{1}\tan\left(3\alpha_{2}\right)\tan\left(3\alpha_{3}\right)\right)}{\tan\left(\alpha_{1}\right)\tan\left(\alpha_{2}\right)\tan\left(\alpha_{3}\right)}\right|$

$=\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(\theta\right)\tan\left(\dfrac{\theta}{3}\right)\tan\left(\dfrac{\theta}{9}\right)}{\tan\left(\dfrac{\theta}{3}\right)\tan\left(\dfrac{\theta}{9}\right)\tan\left(\dfrac{\theta}{27}\right)}\right|=\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(\theta\right)}{\tan\left(\dfrac{\theta}{27}\right)}\right|$

$=\log_{(2+\sqrt{3})}\left|\dfrac{\tan\left(\dfrac{9\pi}{4}\right)}{\tan\left(\dfrac{9\pi}{4.27}\right)}\right|=\log_{(2+\sqrt{3})}\left|\dfrac{1}{\tan\left(\dfrac{\pi}{12}\right)}\right|-\log_{(2+\sqrt{3})}\left|2+\sqrt{3}\right|=1$

14.&15. The function f is increasing

$f'\left(x\right)=3x^{2}-24x+a\geq 0\Rightarrow\Delta\leq 0\Rightarrow a\geq 48$

$fofof\left(3\right)=3\Rightarrow f\left(3\right)=3$

$fofofof\left(4\right)=4\Rightarrow f\left(4\right)=4$

$\Rightarrow a=48,b=-60$

$f\left(7\right)=31$

16.&17.

Given two lines

$\bar{r}=\bar{a}+t\bar{b}$

$\bar{r}=\bar{c}+s\bar{d}$

x_2 is the foot of the perpendicular drawn from x_1 on to the second line. Again x_3 is the foot of the perpendicular drawn from x_2 on to the first line. This process is repeated so that the point of intersection of two lines is obtained.

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PHYSICS

18. A light ray from air $(\mu = 1)$ is incident tangentially on a sphere of radius 2R.
Using Snell's law at the first interface $(\mu_1 = 1 \rightarrow \mu_2 = 2)$, the refracted angle is 30^0
At the second interface $(\mu_2 = 2 \rightarrow \mu_3 = 4)$, the ray bends further.
Due to exceeding the critical angle (30^0) at the inner boundary $(\mu_2 = 4 \rightarrow \mu_2 = 2)$, total internal reflection (TIR) occurs.
After multiple reflections, the net deviation suffered by the light ray is 240^0.

19. $P_0V_0 = nRT_M$

$2P_0V_0 = nRT_N$

$$\Delta Q_{MN} = nC_P\left(T_M - T_N\right) = nC_P\frac{P_0V_0}{nR} = 200\,\text{J}$$

$9P_0V_0 = nRT_0$

$$(\Delta U)_{OM} = nC_V\left(T_0 - T_M\right) = 900\,\text{J}$$

$$\frac{C_P}{C_V} = \gamma = \frac{16}{9}$$

20. Energy of light

$$E_1 = \frac{1240}{620} = 2\text{eV}\quad E_2 = \frac{1240}{496} = 2.5\text{eV}$$

$$E_3 = \frac{1240}{310} = 4\text{eV}$$

No. of photons capable to emit electrons per second

$$= \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{2.5\text{e}} + \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{4\text{e}} = \frac{83.2\times 10^{-6}}{\text{e}}$$

No. of electrons emitted per second $= \dfrac{83.2\times 10^{-6}}{2\times \text{e}}$

Photo current $= ne = 41.6\times 10^{-6}$ ampere

21. $$R = \frac{24}{8} = 3$$

$$4.8 = \frac{24}{\sqrt{(3)^2 + \omega^2L^2}} \Rightarrow L = 0.08$$

$$P = V_{\text{rms}}i_{\text{rms}}\cos\phi = 24 \text{ watt}$$

22. $$mgL\times\frac{4}{5} = mgx$$

$$x = \frac{4L}{5} \qquad \cos\theta = \frac{L-x}{L} = \frac{1}{5}$$

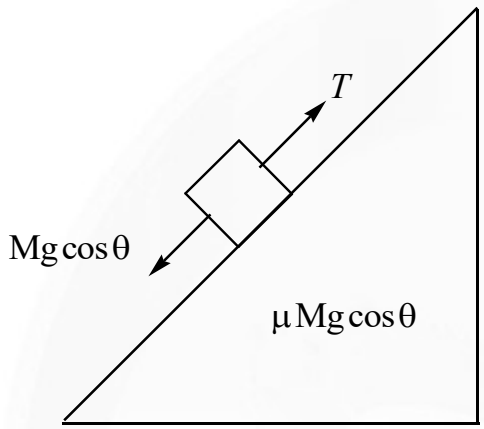
$$mgL\left(e^2\right)^n = mg\frac{64L}{125} \qquad \left(\frac{4}{5}\right)^n = \frac{64}{125}$$

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23. $V=\sqrt{\frac{F}{\mu}}=\frac{10}{(\mu_0+\alpha x)^{1/2}}\Rightarrow\frac{dx}{dt}=\frac{10}{(\mu_0+\alpha x)^{1/2}}$

$\Rightarrow\int_0^{\ell}(\mu_0+ax)^{1/2}dx=\int_0^{t}10dt \qquad t=\frac{1}{10}\times\int_0^{t}(\mu_0+\alpha x)^{1/2}dx$

24. When friction force between block and inclined plane is zero, tension $=mg\sin\theta$



So elongation $=\frac{mg\sin\theta}{k}$

When block m starts to move

$T=mg\sin\theta+\mu mg\cos\theta$.

Elongation in spring $=\frac{mg\sin\theta+\mu mg\cos\theta}{k}$

P.E. stored in spring $=\frac{1}{2}kx^2=\frac{m^2g^2(\sin\theta+\mu\cos\theta)^2}{2k}$

By work energy theorem $W_F=\Delta U+\Delta K.E.$

So $W_F=\frac{1}{2}kx^2$ s Δ K.E. $=0$

25. The forces acting on the block along the plane are component of mg along x-axis and friction force along the tangential direction.

$F_{gr(x)}=mg\sin\alpha$

$f_r = \mu mg\cos\alpha$

$= (\tan\alpha)mg\cos\alpha$

$= mg\sin\alpha$

and $= (mg\sin\alpha)\cos\phi - mg\sin\alpha$

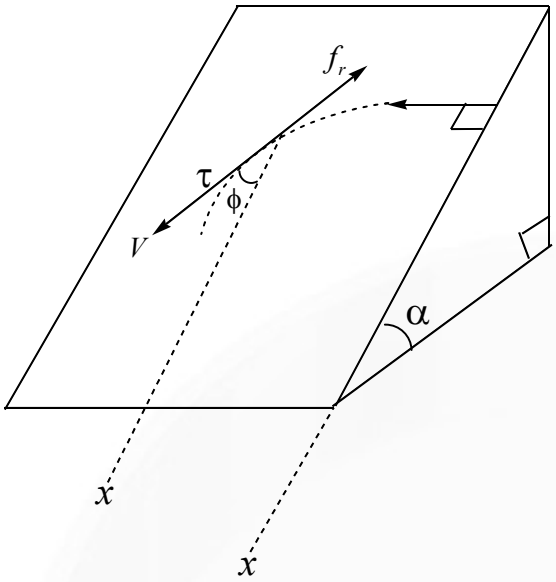
$= mg\sin\alpha(\cos\phi-1)$

and $ma_x = mg\sin\alpha - mg\sin\alpha\cdot\cos\phi$

$= mg\sin\alpha(1-\cos\phi)$

$\therefore a_t=-a_k$ or $\frac{dv}{dt}=-\frac{dv_x}{dt}\therefore dv=-dv_x$

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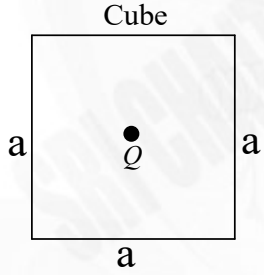


Now, when $\phi = \frac{\pi}{2}, v = 64 \therefore c = 64$.

$\therefore v(1+\cos\phi) = 64 \quad \therefore v = \frac{64}{(1+\cos\phi)}$

the course of time $\phi \to 0 \therefore v = \frac{64}{2} = 32$.

26. Apply Gauss Law, $\phi = \frac{1}{6}\left(\frac{Q}{\varepsilon_0}\right)$



$F = \sigma\phi$.

27. $dN = \frac{N}{\left(\frac{\pi R}{2}\right)}(Rd\theta) \quad B = \int_0^{\frac{\pi}{2}} \frac{\mu_0 i 2N \times d\theta \times R^2 \sin^2\theta}{\pi \times 2R^3} = \frac{\mu_0 iN}{4R}$

28. For parallel process $t_{1/2} = \frac{1}{10} hr$

Total number of half lives $= \frac{1/2}{1/4} + \frac{1}{1/6} + \frac{1/2}{1/10} \therefore n + 5 = 13 \quad = 13 \quad n = 8$

29. $R = \frac{v^2}{a_N} \quad a_N = \frac{2(m+M)}{M} g \qquad$ And $\frac{1}{2}mv^2 = \frac{M}{m+M} mgR$

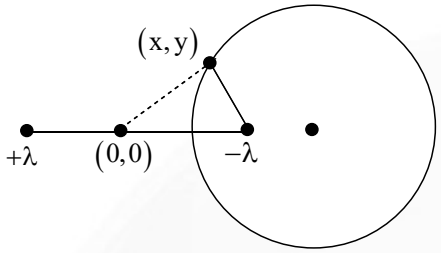
So $r = \left[\frac{M}{m+M}\right]^2 R \qquad r = 4\ cm$

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30. $V=\frac{\lambda}{4\pi\varepsilon_0}\ln\left[(a+x)^2+y^2\right]-c-\frac{\lambda}{4\pi\varepsilon_0}\ln\left[(a-x)^2+y^2\right]+c$

$(a+x)^2+y^2=2(a-x)^2+2y^2 \qquad \Rightarrow (x-3a)^2+(y-0)^2=(2\sqrt{2}a)^2$

Centre of circle$(3a,0)$ $\quad R=2\sqrt{2}a$



31 & 32

$2\mu d=(2n+1)\lambda/2$ for constructive interference $\lambda=\frac{4\mu d}{2n+1}$

$\lambda=4\mu d,\frac{4\mu d}{3},\frac{4\mu d}{5},\frac{4\mu d}{7}\cdots\cdots$ As $n\uparrow\ \lambda\uparrow$

$400=\frac{4\mu d}{2n+1}\Rightarrow 4\mu d=400(2n+1)$

$\frac{400}{2(n-1)+1}<1150\Rightarrow 30n>31\ n>\frac{31}{30}\ n=2,3,\cdots\cdots$

$\frac{4\mu d}{2(n-2)+1}>1150\Rightarrow n<\frac{77}{30}\ n=0,1,2\cdots$

$n=2$ (common)

$\frac{4\times1\times d}{5}=400\ d=500\,\text{nm}.$

$\Delta L=L\propto T\Rightarrow T=\frac{d}{L\propto}=3.1^0C$

$\lambda=666.67\,\text{nm}$

33. $T_B=\frac{P_B}{P_A}T_A=2T_0$

$(2T_0)V_0^{\frac{13}{9}-1}=T_C\cdot(2V_0)^{\frac{13}{9}-1}$

$\Rightarrow\ T_C=2^{5/9}T_0$

34. Heat is absorbed during the process AB and BC

$Q_{AB}=\frac{3}{2}R(T_B-T_A)=\frac{3}{2}RT_0$

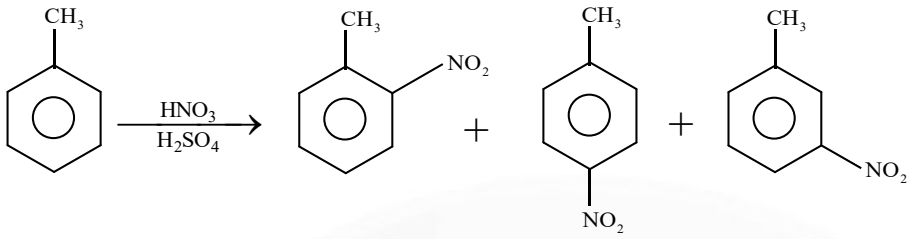
$Q_{BC}=\left(\frac{3}{2}R-\frac{9}{4}R\right)(T_C-T_B)=\frac{3}{4}R\times(2-2^{5/9})T_0$

$Q_{abs}=\frac{3}{4}(4-2^{5/9})RT_0\cong\frac{15}{8}RT_0\left[2^{5/9}\cong1.5\right]$

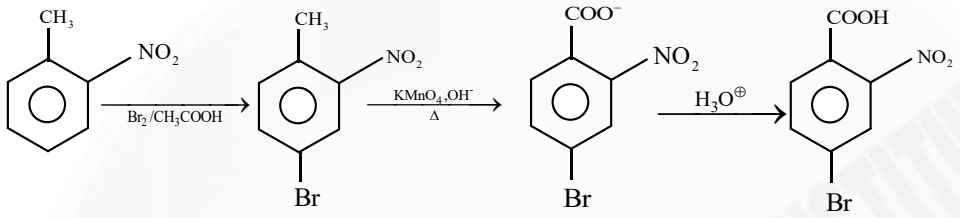
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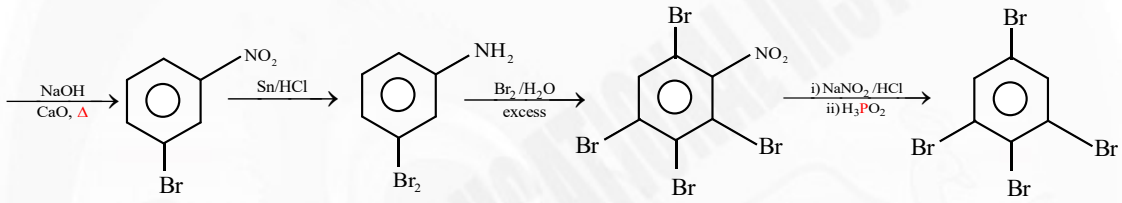
CHEMISTRY

35.



P(59%) Q(37%) R(4%)





36. (A) Aldehyde carbon of p-nitro benzaldehyde is more electrophilic and readily attacked by OH^-.

37. $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$

$= 96 + 430 - 126 = 400\ \Omega^{-1}cm^2\ mol^{-1}$

$\Lambda_m^\circ(CH_3COOH) = \frac{k \times 1000}{M} = \frac{5 \times 10^{-5} \times 1000}{10^{-3}} = 50\ Scm^2\ mol^{-1}$

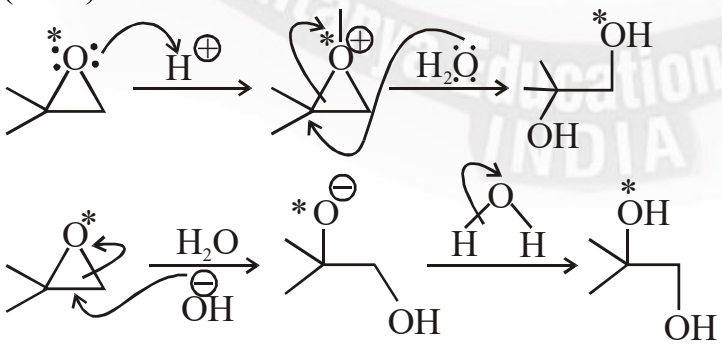
$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{50}{400} = \frac{1}{8}$

$\pi = (1+\alpha)CRT = \frac{9}{8} \times 10^{-3} RT = 1.125 \times 10^{-3} RT$

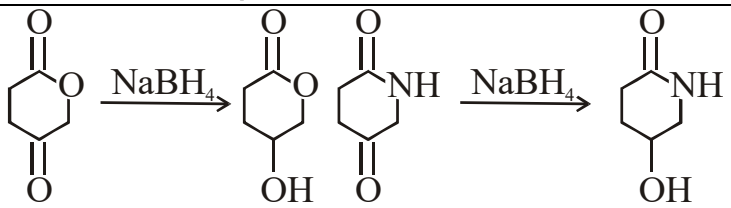
38.

39. (ABD) Formal charge on Sulfur (S) is 0 in SF_2, SF_4 *and* SF_6

40. (ABC)



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41. (ABCD)

Eq of H_2O_2 = Eq of $KMnO_4$

$20 \times x = 50 \times 0.2$

$x = 0.5N$

$V.f$ of H_2O_2 is 2

$\Rightarrow Molarity = \frac{0.5}{2} = 0.25$

Volume strength = Molarity $\times 11.2 = 2.8$

42. O_2^{-2} & $K_2[PtCl_4]$ are diamagnetic rest all are paramagnetic.

43. $$\Delta T = \frac{1000K_f' \times w}{W \times m}$$

In C_6H_6: $$1.28 = \frac{1000 \times 5.12 \times w}{m_N \times 100} \quad .....(1)$$

In H_2O: $$1.40 = \frac{1000 \times 5.12 \times w}{m_{exp} \times 100} \quad .....(2)$$

(Since given that solute behave as normal in C_6H_6 and dissociates in water)

By Eqs, (1) and (2)

$$\frac{m_N}{m_{exp}} = \frac{1.40}{1.28} \times \frac{5.12}{1.86} = 3.01$$

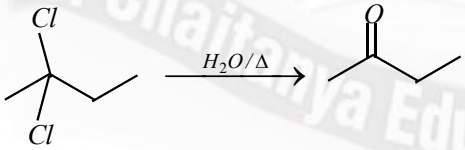
$\therefore$ $i = 3.01 \approx 3.0$

44. For an electrochemical cell,

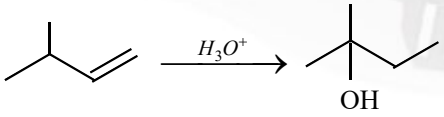
$\Delta G° = -nFE^0_{cell} = -2 \times 96485 \times 1.1$

$= -212267J$ $\quad -53x = -212kJ \Rightarrow \boxed{x = 4}$

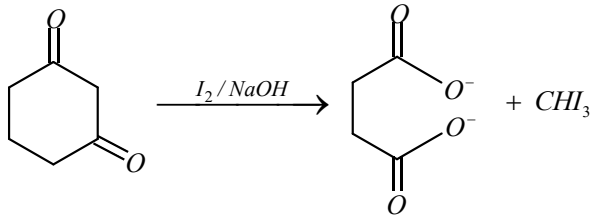
45.

(i) 

(can give Iodoform)

(ii) 

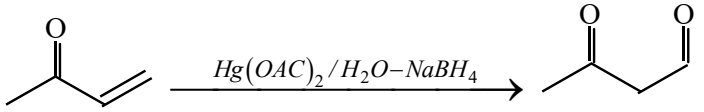
(does not give iodoform)

(iii) 

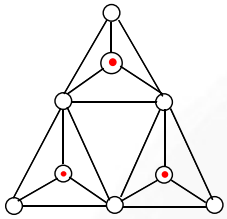
+ CHI_3

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(iv) β – keto ester does not give Iodoform

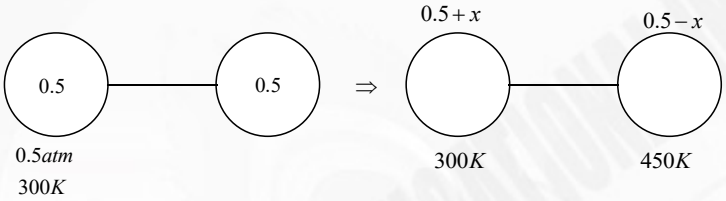
(v)  (can give iodoform)

(vi) $Ph-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{I_2/NaOH}$ (Can give iodoform)

46.  x = 3, y=6

47. E^0_{SRP} for Li^+/Li is more negative than Na^+/Na. But Na is mote reactive than Li.

48 & 49.



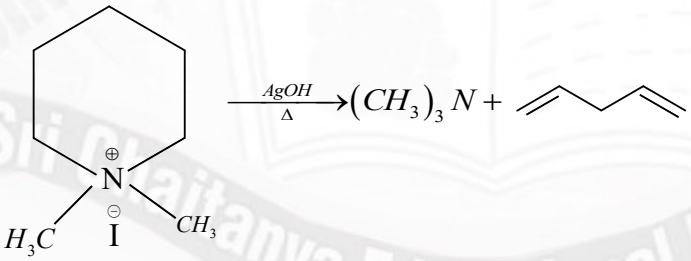
$0.5+x(300)=(0.5-x)450$

$1+2x=1.5-3x$

$x=\frac{0.5}{5}=.01 mole$

$\frac{p_1}{n_1}=\frac{p_2}{n_2}\Rightarrow\frac{0.5}{0.5}=\frac{p_2}{0.6}=p_2=0.6\,atm$

50.

51. 

Molar mass of alkene, $R=68 \therefore \frac{R}{10}=\frac{68}{10}=6.8$

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